# Semester Wrap-Up and Thesis Progress

## Barrier Function-Based Navigation for Precision Agriculture

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# Agenda

1. Project overview
2. What was achieved
3. Current status
4. Next steps
5. Closing

# Project Overview

- Goal: Safe 3D navigation for micro-UAVs in greenhouse-like environments.
- Core components:
  - Monocular SLAM for onboard localization.
  - Generative planning to exploit structured layouts.
  - 3D Control Barrier Functions for safety [1].
  - Baseline PD controller (other options will be explored later).
  - Hardware and robustness (prop-guards, disturbance rejection).



# System Architecture

- **Onboard** Crazyflie:
  - IMU, barometer, optical flow deck, monocular camera.
  - Low-level stabilization and sensor acquisition.
- **Offboard** workstation:
  - SLAM, generative planner, CBF safety filter.
  - Visualization, logging, data analysis.
- **OptiTrack**: ground truth for development and benchmarking.

# Progress Aligned with Project Plan

- **Literature review**: Updated (CBFs, SLAM, greenhouse UAV applications).
- **Scenario & setup**: Virtual and real greenhouse-like environments defined.
- **Robot preparation**: Crazyflie configured with markers, decks, and data link.
- **Thesis writing**: Proposal delivered; chapter structure initiated.

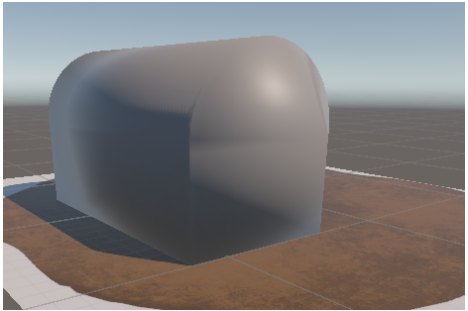
# Experimental Platform

- Reliable Crazyflie–PC communication (radio + logging).
- Closed-loop flights using OptiTrack feedback.
- Baseline PD control validated for hover and waypoints.
- Repeatable routines: hover tests, step responses, short trajectories.



# Experimental Setup

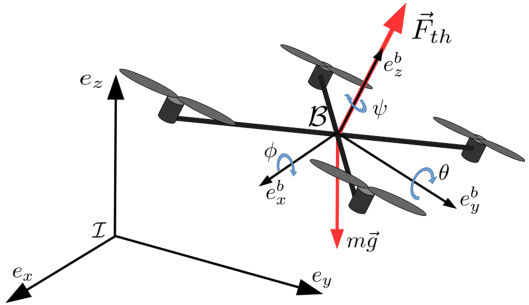
- **Hardware:** Crazyflie 2.1, OptiTrack, RTX workstation.
- **Environments:**
  - Mocap arena simulating greenhouse corridors.
  - Unity greenhouse (transitioning to Gazebo).





# Modeling and Sensors

- 6-DoF quadrotor model implemented.
- Sensors integrated:
  - IMU, barometer, optical flow, monocular camera.
  - OptiTrack for benchmarking only.
- Infrastructure supports:
  - 3D CBF design and evaluation.
  - Low-level controller testing (PD $\rightarrow$ PID $\rightarrow$ possibly MPC).



# MLP-Based Pose Estimation

- Dataset: IMU + barometer + camera aligned with OptiTrack.
- MLP trained to estimate 6-DoF pose from onboard sensors.
- Good offline reconstruction; used mainly for insight.
- Future: inform SLAM or act as fallback estimate.

# MLP Results

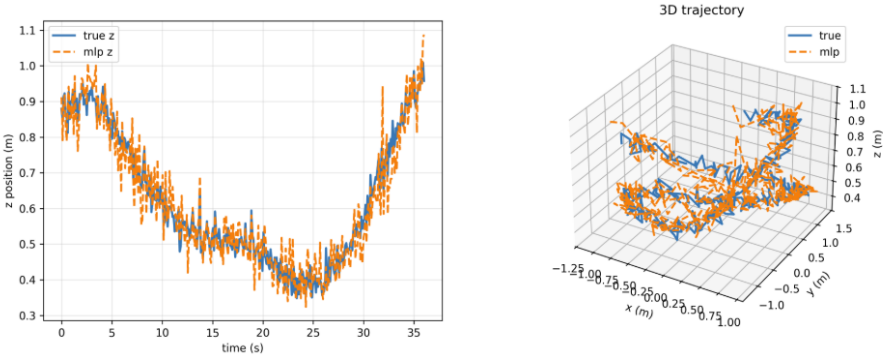



Figure: Ground-truth vs. MLP-predicted trajectories.

# Published Results

**Energy-Aware Hybrid Event-Triggered Control for Micro-Drones**

**Published in:** IMAV 2025 [5]

**Summary:**

- Hybrid event-triggering strategy reduces communication without sacrificing control performance.
- Adaptive threshold + safety timer lowers total energy use compared to periodic control.
- Demonstrated 25–40% packet savings with minimal tracking error increase.

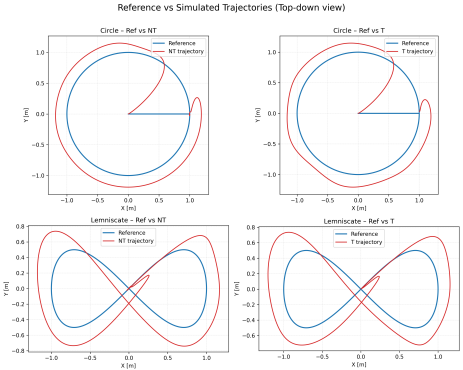


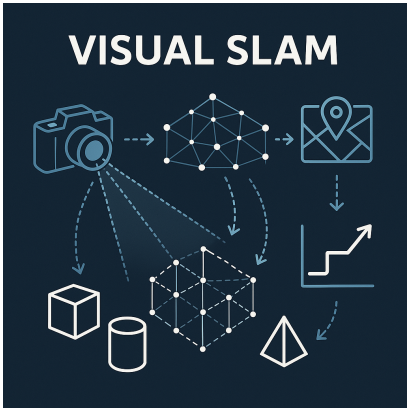

Figure: Ground-truth vs. MLP-predicted trajectories.

# Summary

- First-year goals met:
  - Literature review, robot preparation, experimental setup ✓.
  - Proposal and initial thesis structure ✓.
- Platform ready for next phases:
  - Robust SLAM in greenhouse-like scenes ✓.
  - 3D CBF safety filter ✓.
  - Generative path planning ✓.

# Next Steps: Perception

- Strengthen monocular SLAM in low-texture environments.
- Characterize drift and failure modes.
- Integrate SLAM with MLP and other onboard sensors [2].



# Next Steps: Planning and Safety

- Develop 3D CBF safety constraints for obstacles and greenhouse structure.
- Evaluate CBF-QP paired with PD/PID/MPC controllers.
- Prototype diffusion-based or sampling-based generative planner [3].
- Combine planner + CBF for safe navigation [4].



# Documentation and Publications

- Prepare targeted publications:
  - 3D CBF extension for aerial navigation.
  - Generative planning + safety layer.
  - Learned localization insights.

# Take-Home Message

- Foundation for the thesis is in place: robot, environment, baseline control, and data.
- Validated early components: PD control, SLAM pipeline, MLP estimator.
- Next objective: integrate perception, planning, and safety for autonomous navigation.

**Thank you for your attention.**

# Bibliography


[1] Aaron D. Ames et al. "Control Barrier Function Based Quadratic Programs for Safety Critical Systems". In: *IEEE Transactions on Automatic Control* 62.8 (2017), pp. 3861–3876. DOI: 10.1109/TAC.2016.2638961.

[2] Cesar Cadena et al. "Past, Present, and Future of Simultaneous Localization and Mapping: Toward the Robust-Perception Age". In: *IEEE Transactions on Robotics* 32.6 (2016), pp. 1309–1332. DOI: 10.1109/TRO.2016.2624754.

[3] João Carvalho et al. "Motion Planning Diffusion: Learning and Planning of Robot Motions with Diffusion Models". In: *arXiv preprint arXiv:2308.01557* (2023). URL: https://arxiv.org/abs/2308.01557.

[4] Brian Ichter, James Harrison, and Marco Pavone. "Learning Sampling Distributions for Robot Motion Planning". In: *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2018, pp. 7087–7094. DOI: 10.1109/ICRA.2018.8460692.

[5] Jesús O. Rodríguez-Ramírez et al. "Energy-Aware Hybrid Event-Triggered Control for Micro-Drones". In: *International Micro Air Vehicle Conference (IMAV)*. Published version. 2025.